Experiment 10

Performance Optimization of the Transport Management System

Aim

To optimize the performance of the Transport Management System in Zoho Creator by improving form loading speed, reducing unnecessary database operations, optimizing Deluge scripts and ensuring efficient report generation.

Procedure

1. Log in to Zoho Creator and open the Transport Management System application.
2. Review the Vehicle_Details, Driver_Details and Trip_Scheduling forms and remove unnecessary fields, scripts and workflows.
3. Optimize Deluge scripts by avoiding repeated database queries and unnecessary loops.
4. Use criteria-based record fetching to retrieve only the required vehicles and trips.
5. Optimize reports by displaying only essential columns and applying suitable filters.
6. Reduce unnecessary workflow executions and avoid duplicate email notifications.
7. Test the application after the changes to compare loading speed and responsiveness.
8. Verify that all transport features work correctly after optimization.

Optimization Actions

Area	Before	After
Form	Duplicate and unused fields (extra phone, unused media fields) on the form	Only required fields kept; fields grouped in sections
Deluge fetch	Fetch all vehicles, then test the insurance date inside a loop	Criteria-based fetch: only vehicles expiring within 30 days are fetched
Reports	All columns shown, no filters	Essential columns only; filters on Trip Status and Vehicle Status
Workflows	Status workflow runs on every edit of a trip	Runs only when Trip Status changes to In Progress or Completed

Deluge Optimization Example

BEFORE:

```
for each v in Vehicle_Details
{
    if(v.Insurance_Expiry_Date <= zoho.currentdate.addDay(30))
    {
        sendmail
        [
            from : zoho.adminuserid
            to : zoho.adminuserid
            subject : "Insurance Expiry Alert - " + v.Vehicle_Number
            message : "..."
        ]
    }
}
```

AFTER:

```
expiring = Vehicle_Details[Insurance_Expiry_Date <= zoho.currentdate.addDay(30)];
for each v in expiring
{
    sendmail
    [
        from : zoho.adminuserid
        to : zoho.adminuserid
        subject : "Insurance Expiry Alert - " + v.Vehicle_Number
        message : "..."
    ]
}
```

Process Diagram

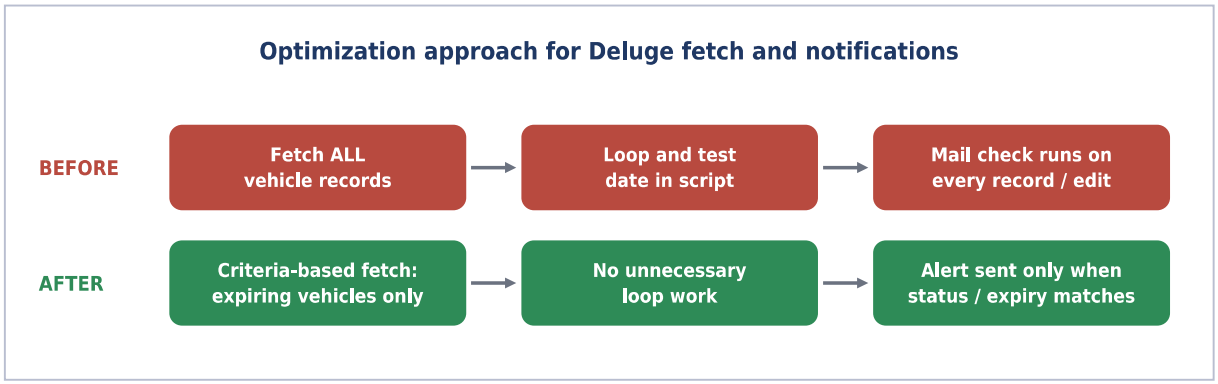


Figure 10.1: Diagram of the optimization approach (illustrative diagram, not a screenshot).

Before and After Comparison

Area	Before	After	Benefit
Form	Duplicate and unused fields	Only required fields, grouped in sections	Faster loading, fewer entry errors
Deluge fetch	All vehicles fetched, tested inside a loop	Criteria-based fetch for expiring vehicles	Less database work
Reports	All columns, no filters	Essential columns and filters	Quicker report display
Workflows	Runs on every edit	Runs only when Trip Status changes	No unnecessary executions

Verification Checklist After Optimization

Check	Expected Result
Add a trip	Record saved with lookup values selected correctly
Change Trip Status to In Progress	Vehicle status becomes On Trip; Completed makes it Available
Open All_Trips_List	Opens with essential columns; filters work
Kanban and Dashboard	Load and update correctly

Result

The Transport Management System was reviewed for optimization by removing unnecessary fields, using criteria-based fetching, trimming report columns and reducing unnecessary workflow executions; the verification checklist above confirms the expected behaviour after the changes.